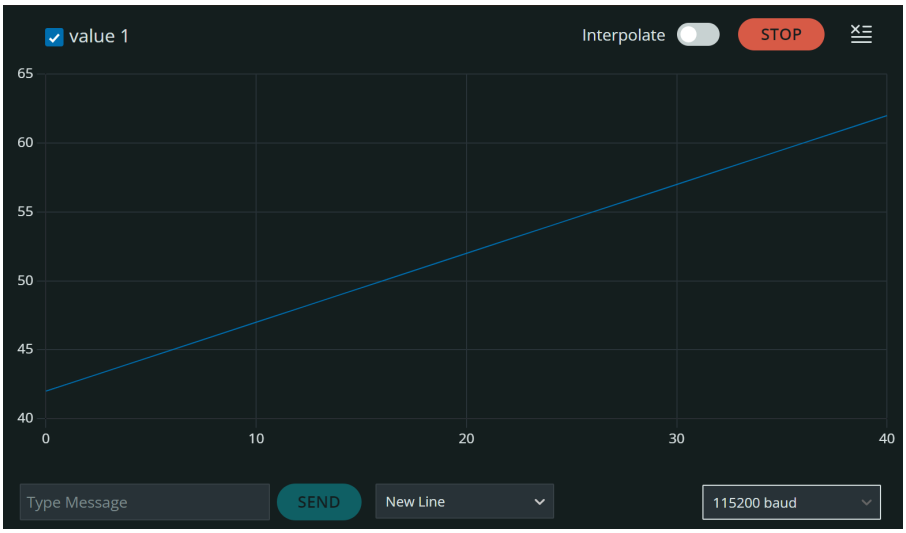
Cliff Pham 2360594 Lab 2 446

GitHub Link : Here

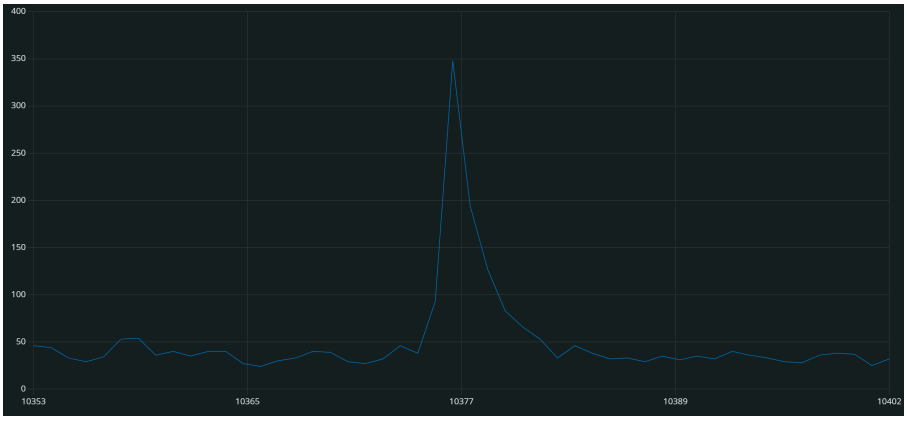
Task 4

image showing counter in serial monitor

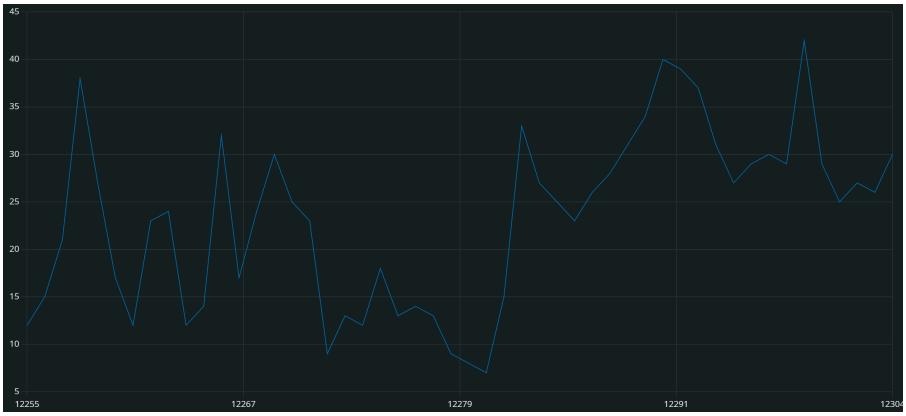


serial plotter

Task 5



quiet mic plot

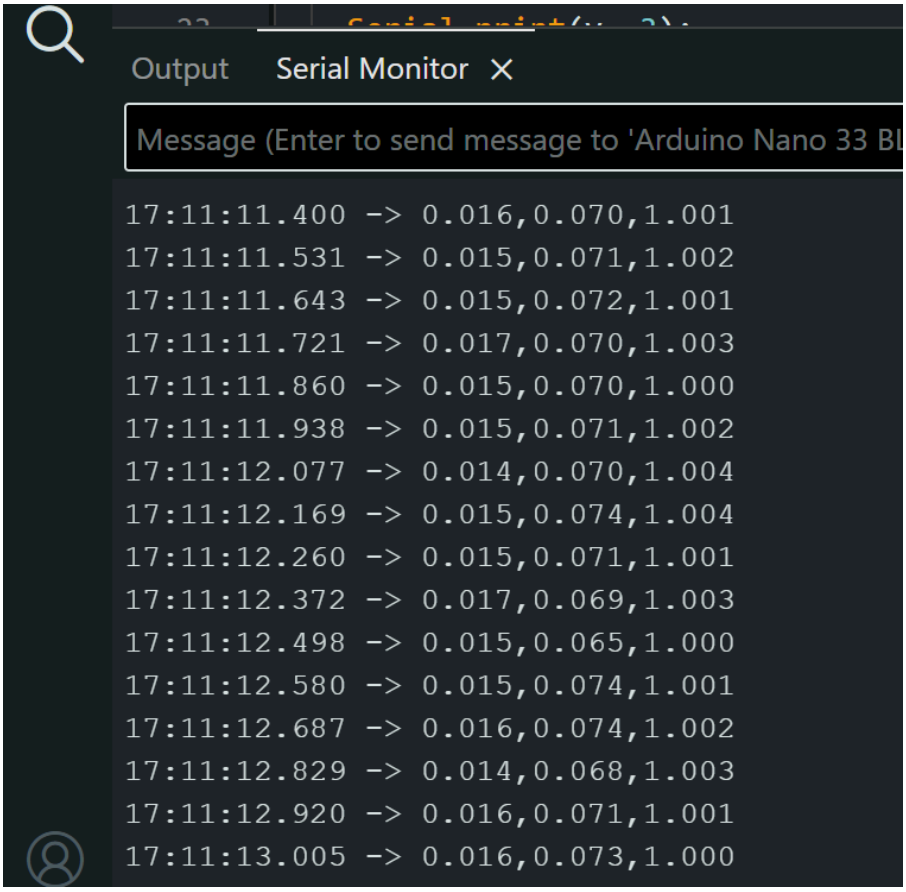


background noise plot

The largest change were sudden loud noises increasing amplitude of serial plot.

Task 6:

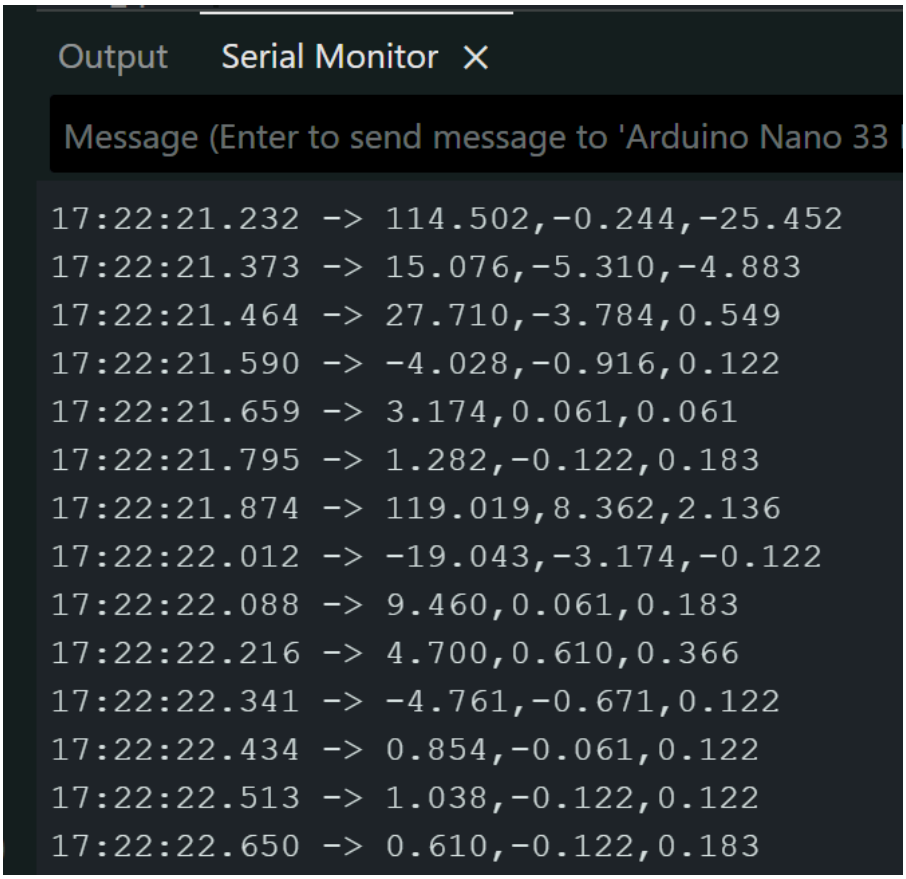
accelerometer serial monitor



accelerometer serial plotter

turning board upside down and back up repeatedly causes all value to spike very fast

Task 6B:

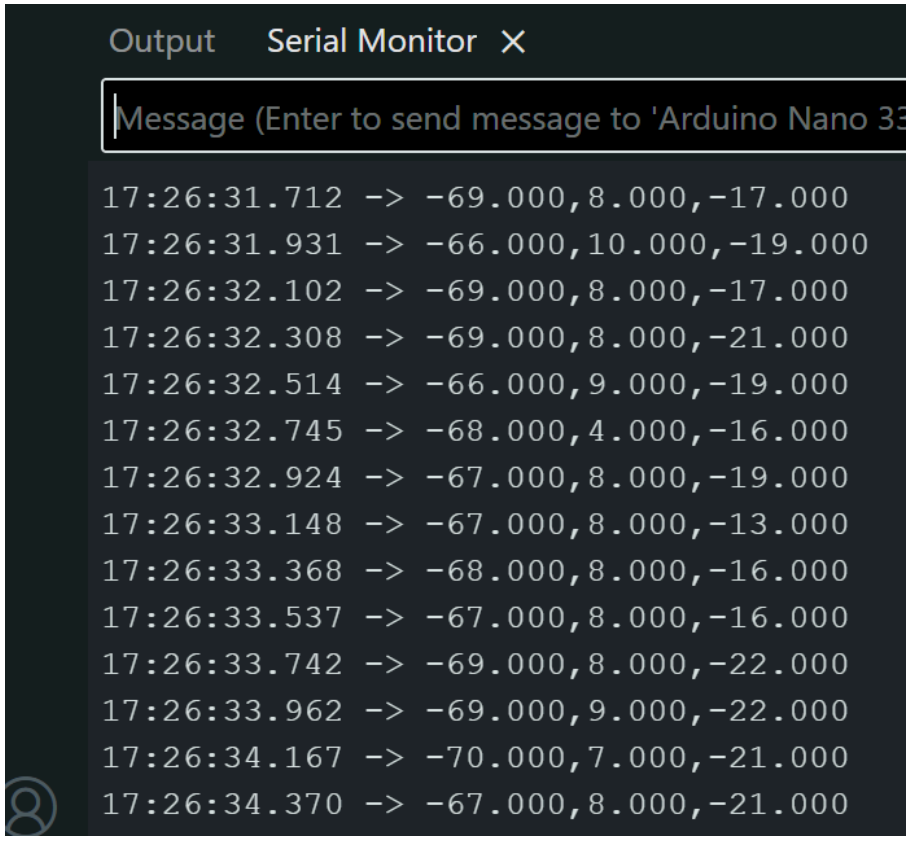


gyroscope serial monitor

gyroscope serial plot

holding board vertically and spinning it any direction fast changes value 1 drastically

Task 6C:



serial montior

serial plotter

Holding next to laptop device will change the values in the serial plotter drastically, or hovering my keys near the board.

task 7:

```
17:33:35.078 -> Temperature (C): 25.81 | Humidity (%): 31.09
17:33:36.158 -> Temperature (C): 25.82 | Humidity (%): 30.28
17:33:37.227 -> Temperature (C): 25.85 | Humidity (%): 31.13
17:33:38.306 -> Temperature (C): 25.85 | Humidity (%): 31.11
17:33:39.344 -> Temperature (C): 25.86 | Humidity (%): 32.99
17:33:40.433 -> Temperature (C): 25.87 | Humidity (%): 33.12
17:33:41.499 -> Temperature (C): 25.87 | Humidity (%): 32.35
17:33:42.603 -> Temperature (C): 25.87 | Humidity (%): 31.30
17:33:43.672 -> Temperature (C): 25.90 | Humidity (%): 31.09
17:33:44.713 -> Temperature (C): 25.91 | Humidity (%): 31.70
```

stable condition

```
17:34:06.175 -> Temperature (C): 25.94 | Humidity (%): 27.76
17:34:07.247 -> Temperature (C): 25.91 | Humidity (%): 27.69
17:34:08.350 -> Temperature (C): 25.94 | Humidity (%): 28.01
17:34:09.418 -> Temperature (C): 26.00 | Humidity (%): 31.03
17:34:10.467 -> Temperature (C): 26.70 | Humidity (%): 46.60
17:34:11.572 -> Temperature (C): 26.92 | Humidity (%): 57.07
17:34:12.635 -> Temperature (C): 27.76 | Humidity (%): 58.41
17:34:13.718 -> Temperature (C): 27.93 | Humidity (%): 58.20
17:34:14.781 -> Temperature (C): 28.04 | Humidity (%): 58.18
17:34:15.862 -> Temperature (C): 27.82 | Humidity (%): 62.54
17:34:16.908 -> Temperature (C): 27.14 | Humidity (%): 52.00
17:34:17.998 -> Temperature (C): 27.36 | Humidity (%): 60.90
17:34:19.079 -> Temperature (C): 27.40 | Humidity (%): 64.83
17:34:20.147 -> Temperature (C): 27.28 | Humidity (%): 61.99
17:34:21.232 -> Temperature (C): 27.31 | Humidity (%): 60.18
17:34:22.301 -> Temperature (C): 27.23 | Humidity (%): 56.24
17:34:23.393 -> Temperature (C): 28.29 | Humidity (%): 75.59
17:34:24.422 -> Temperature (C): 27.93 | Humidity (%): 78.81
17:34:25.523 -> Temperature (C): 27.80 | Humidity (%): 79.90
```

hand over board condition

Task 8:

```
17:38:59.738 -> Pressure (kPa): 101.679 | Temperature (C) :26.13
17:39:00.747 -> Pressure (kPa): 101.674 | Temperature (C) :26.13
17:39:01.734 -> Pressure (kPa): 101.670 | Temperature (C) :26.12
17:39:02.798 -> Pressure (kPa): 101.669 | Temperature (C) :26.12
17:39:03.801 -> Pressure (kPa): 101.673 | Temperature (C) :26.12
17:39:04.833 -> Pressure (kPa): 101.675 | Temperature (C) :26.12
17:39:05.849 -> Pressure (kPa): 101.675 | Temperature (C) :26.12
17:39:06.850 -> Pressure (kPa): 101.676 | Temperature (C) :26.12
17:39:07.861 -> Pressure (kPa): 101.677 | Temperature (C) :26.11
17:39:08.895 -> Pressure (kPa): 101.679 | Temperature (C) :26.11
17:39:09.882 -> Pressure (kPa): 101.679 | Temperature (C) :26.11
17:39:10.925 -> Pressure (kPa): 101.678 | Temperature (C) :26.11
17:39:11.944 -> Pressure (kPa): 101.681 | Temperature (C) :26.12
```

serial monitor for pressure sensor

pressure reading was mostly stable as it is hard ot change environmental pressure.

task 9a:

```
18:07:41.970 -> 241
18:07:42.205 -> Proximity :
18:07:42.205 -> 242
18:07:42.376 -> Proximity :
18:07:42.376 -> 243
18:07:42.613 -> Proximity :
18:07:42.613 -> 242
18:07:42.787 -> Proximity :
18:07:42.787 -> 242
18:07:42.992 -> Proximity :
18:07:42.992 -> 241
18:07:43.212 -> Proximity :
18:07:43.212 -> 240
18:07:43.394 -> Proximity :
18:07:43.394 -> 242
18:07:43.616 -> Proximity :
18:07:43.616 -> 241
```

hand far away

```
18:08:26.850 -> 0
18:08:27.070 -> Proximity :
18:08:27.070 -> 0
18:08:27.243 -> Proximity :
18:08:27.243 -> 0
18:08:27.462 -> Proximity :
18:08:27.462 -> 10
18:08:27.696 -> Proximity :
18:08:27.696 -> 0
18:08:27.854 -> Proximity :
18:08:27.854 -> 0
18:08:28.057 -> Proximity :
18:08:28.057 -> 0
18:08:28.290 -> Proximity :
18:08:28.290 -> 0
18:08:28.464 -> Proximity :
18:08:28.464 -> 0
```

hand close

As my hand is farther away, the proximity value increases and it decreases as my hand gets closer to the board.

Task 9B:

```
Output    Serial Monitor ×
Message (Enter to send message
18:14:50.978 -> DOWN
18:14:51.048 -> UP
18:14:51.659 -> RIGHT
18:14:51.738 -> LEFT
18:14:52.476 -> RIGHT
18:14:52.555 -> LEFT
18:14:52.727 -> RIGHT
18:14:52.963 -> LEFT
18:14:53.120 -> RIGHT
18:14:53.198 -> LEFT
18:14:54.072 -> RIGHT
```

left and right and up and down were recognized reliably if move hand back and forth really fast

Task 9C:

```
18:18:03.498 -> 53,46,35,124
18:18:03.903 -> 52,45,34,122
18:18:04.308 -> 52,45,34,122
18:18:04.758 -> 52,45,34,122
18:18:05.147 -> 52,44,34,122
18:18:05.551 -> 52,44,34,122
18:18:05.965 -> 52,44,34,122
18:18:06.333 -> 52,44,34,122
18:18:06.782 -> 51,43,33,119
18:18:07.143 -> 51,43,33,119
18:18:07.547 -> 49,41,31,114
```

base condition

```
18:18:26.619 -> 7,5,4,15
18:18:27.025 -> 6,3,3,12
18:18:27.403 -> 7,4,3,13
18:18:27.814 -> 7,4,3,13
18:18:28.223 -> 7,4,3,13
18:18:28.630 -> 6,3,3,11
18:18:29.013 -> 6,3,2,11
```

showing yellow color

yellow color produced clearest difference because I assume it is the brightest compared to the current environmental lighting conditions.

Task 10

QUIET_BRIGHT_STEADY_FAR:

```
00:15:48.927 -> flags,sound=0,dark=0,moving=0,near=0
00:15:48.927 -> state,QUIET_BRIGHT_STEADY_FAR
00:15:49.201 -> raw, mic=17,clear=569,motion=0.02,prox=242
00:15:49.238 -> flags,sound=0,dark=0,moving=0,near=0
00:15:49.238 -> state,QUIET_BRIGHT_STEADY_FAR
00:15:49.549 -> raw, mic=17,clear=573,motion=0.01,prox=244
00:15:49.549 -> flags,sound=0,dark=0,moving=0,near=0
00:15:49.549 -> state,QUIET_BRIGHT_STEADY_FAR
```

NOISY_BRIGHT_STEADY_FAR:

```
00:17:06.251 -> raw, mic=640,clear=58,motion=0.01,prox=245
00:17:06.251 -> flags,sound=1,dark=0,moving=0,near=0
00:17:06.251 -> state,NOISY_BRIGHT_STEADY_FAR
```

QUIET_DARK_STEADY_NEAR:

```
00:14:29.011 -> flags,sound=0,dark=1,moving=0,near=1
00:14:29.011 -> state,QUIET_DARK_STEADY_NEAR
00:14:29.340 -> raw, mic=9,clear=4,motion=0.00,prox=0
00:14:29.340 -> flags,sound=0,dark=1,moving=0,near=1
00:14:29.340 -> state,QUIET_DARK_STEADY_NEAR
00:14:29.618 -> raw, mic=39,clear=6,motion=0.01,prox=0
00:14:29.618 -> flags,sound=0,dark=1,moving=0,near=1
00:14:29.618 -> state,QUIET_DARK_STEADY_NEAR
```

NOISY_BRIGHT_MOVING_NEAR:

```
00:21:18.777 -> raw, mic=187,clear=4097,motion=2.26,prox=69
00:21:18.809 -> flags,sound=1,dark=0,moving=1,near=1
00:21:18.809 -> state,NOISY_BRIGHT_MOVING_NEAR
```

I selected my thresholds by using the previous tasks as a baseline for what I should set my threshold as, and then changed it accordingly through experimentation to ensure the smart workspace situation classifier works robustly.

Task 11

BASELINE_NORMAL

```
15:06:29.006 -> raw,rh=40.09,temp=22.90,mag=2.00,r=5,g=5,b=5,clear=15
15:06:29.006 -> flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=0
15:06:29.006 -> event,BASELINE_NORMAL
```

BREATH_OR_WARM_AIR_EVENT

```
15:08:38.770 -> raw,rh=59.70,temp=22.79,mag=5.00,r=8,g=7,b=7,clear=23
15:08:38.809 -> flags,humid_jump=1,temp_rise=0,mag_shift=1,light_or_color_change=0
15:08:38.809 -> event,BREATH_OR_WARM_AIR_EVENT
```

MAGNETIC_DISTURBANCE_EVENT

```
15:06:29.581 -> raw,rh=41.09,temp=22.88,mag=3.00,r=5,g=5,b=5,clear=15
15:06:29.581 -> flags,humid_jump=0,temp_rise=0,mag_shift=1,light_or_color_change=0
15:06:29.581 -> event,MAGNETIC_DISTURBANCE_EVENT
```

LIGHT_OR_COLOR_CHANGE_EVENT

```
15:13:31.890 -> raw,rh=38.69,temp=22.30,mag=4.00,r=2,g=5,b=8,clear=15
15:13:31.932 -> flags,humid_jump=0,temp_rise=0,mag_shift=0,light_or_color_change=1
15:13:31.932 -> event,LIGHT_OR_COLOR_CHANGE_EVENT
```

I selected my thresholds similar to how I did in task 10. I used the previous tasks as a starting benchmark threshold, and then I experimentally changed the values based on the current enviornment condition my board was in.